48-Hour Electricity Generation Forecasting
MATH840 | Time Series Final Project

Illia Hrynkiv

What is in the series?

Dataset: ~3.8 years of historical hourly data (N ≈ 33,300 hours).
Patterns: Strict 24-hour (daily) and 168-hour (weekly) seasonality.
Weather Impact: High correlation with temperature and daylight.
Key Anomaly: Nighttime generation drops near 0, causing extreme MAPE sensitivity.

Validation Strategy

Method: 48-hour hold-out validation window (the final 48 hours of train.csv).
Baseline: Seasonal Naive (SNAIVE).
SNAIVE Benchmark Metrics on Hold-out:
RMSE: 120.40
MAPE: 51.09%
Winkler Score (α=0.05): 568.46
Coverage: 100%

Failed Approaches

Prophet (Multiplicative): Failed. RMSE spiked to 797.7, MAPE to 421.9%, and Winkler to 14,975.
SARIMAX: Suboptimal. RMSE 161.4 (worse than the 120.4 baseline).
Unconstrained LightGBM: Overfitted on transition-hour noise.

Building the 95% Interval

Method: Quantile Regression in HistGradientBoostingRegressor ($\alpha = 0.025$ and 0.975).
Interval Ensemble: 50% Quantile Trees + 50% SNAIVE intervals.
Winkler Penalty Post-processing:

- lower_bound = max(0, lower_bound)
- lower_bound <= point_forecast <= upper_bound

Validation Results: Coverage = 100%, Winkler = 571.89.

Final Model & Results

Final Architecture: 60% Regularized LGBM + 40% SNAIVE.

Final Hold-out Metrics vs. Benchmark:

RMSE: 120.40 → 87.08 (-27.6%)

MAPE: 51.09% → 49.50%

Successfully surpassed the baseline threshold for maximum session points